MILITARY SPECIFICATION

FUZE, M539, IMPACT SWITCH ASSEMBLY FOR, PARTS FOR, ASSEMBLING AND PACKING

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Military Specification covers parts for assembling and packing of Fuze, M539, Impact Switch Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods Of
MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging
MIL-B-3106	- Board, Composition, Water-resistant, Solid (For Filler or Cushioning Pads)
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of
MIL-F-50449	- Filler, Sheet Form (For Use In Ammunition Containers)

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin

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PPP-B-601 - Box, Wood, Cleated Plywood
PPP-B-636 - Box, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing And Marking for Shipment of Inert Ammunition Components

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGE DRAWINGS

9276238 - Power Supply, Fuze, M539
9276237 - Impact Switch Assembly, Fuze, M539
9293288 - Projectile, 152MM, HEAT-T-MP, M409A2, Loading Assembly
9294989 - Cartridge, 152MM, HEAT-T-MP, M409A2

FRANKFORD ARSENAL

10534412 - Windshield
10534417 - Insulator Assembly

3. REQUIREMENTS

3.1 General. -Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Impact Switch.-The impact switch shall comply with all requirements specified on dwg. 9276237 and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be free of chips, dirt, grease, rust, burrs, cracks and other foreign material. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.4 Proving Ground.-The applicable loaded projectile shall comply with the following requirements when assembled with an impact switch and tested as specified in 4.5.4.

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3.4.1 Graze Functioning.-The projectile shall detonate "high-order" on initial impact with a gravel pad located 1000 ± 200 meters from the gun muzzle.

3.4.2 Safety.-There shall be no indication of impact switch separation or breakup in the gun bore or in flight prior to impact.

3.5 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 10 each of every component and every assembly. (Where a multiple cavity mold is used in the manufacturing process, a minimum of 5 samples shall be submitted from each mold cavity. Each part shall be identified with its mold cavity).

4.3.2 Inspections to be Performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to all examinations and tests specified in 4.4.2 of this specification and to any or all requirements of the applicable drawings. 5 each of the sample assemblies (2 each if multiple cavity mold is used) shall be temperature conditioned at 180°F ± 10°F for 24 ± 1 hours and then held at room ambient temperature for 4 hours minimum prior to inspection.

4.3.3 Rejection.-See MIL-A-48078 (PA).

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.1	Impact Switch Assembly, Fuze, M539		9276237
			NEXT HIGHER ASSEMBLY
			9293288/9294989

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
			4.4.3.1.2	3.4.2	Visual/Photographic
Critical					
1	Safety*		0.40%	3.2	Gage (Note)
Major A			0.40%	3.2	Gage (note)
101	Runout, forward		4.4.3.1.1	3.4.1	4.5.4.1
102	Runout, aft		100%	3.2	4.5.2
103	Graze Functioning*				
104	Insulation resistance, min,		0.40%	3.2	Gage
Major B			0.40%	3.2	Gage
131	Height		0.40%	3.2	Gage
132	Dimension from shoulder to forward end		0.40%	3.2	Gage
133	Location of forward internal basic diameter		0.40%	3.2	Gage
134	Location of aft internal basic diameter		0.40%	3.2	Gage
135	Height of ribs		0.40%	3.2	Gage
136	Thickness, through ribs		0.40%	3.2	Gage
137	Inside diameter of relief cavities in shoulder		0.40%	3.2	Gage
138	Outside diameter of relief cavities in shoulder		0.40%	3.2	Gage
139	Thread pitch diameter, max		0.40%	3.2	Visual
140	Thread minor diameter, max		0.40%	3.2	Visual/Scale
141	Any feature missing		0.40%	3.2	Gage
142	Excessive flash between ribs				
143	Inside diameter over power supply centering ribs		0.65%	3.2	Gage
Minor					
201	Dimension between ribs				

NOTES: Note: Optional reduced inspection (para. 4.5.3) not applicable

* Proving Ground

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63068 (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	CONTINUATION SHEET				9276237
					NEXT HIGHER ASSEMBLY
					9293288/9294989
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
202	Dimension from bottom of groove to aft edge of switch		0.65%	3.2	Gage
203	Angle at aft edge of switch		0.65%	3.2	Gage
204	Depth of relief cavities		0.65%	3.2	Gage
205	Diameter of spanner holes		0.65%	3.2	Gage
206	Dimension between centers of spanner holes		0.65%	3.2	Gage
207	Dimension from top of groove to shoulder		0.65%	3.2	Gage
208	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63068 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Cup, Impact Switch		9276243
			NEXT HIGHER ASSEMBLY 9276237

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Overall depth		0.40%	3.2	Gage
102	Dimension from outside body radius to inside of nose end		0.40%	3.2	Gage (Note 1)
103	Angle of body between inside and outside radii		0.40%	3.2	Gage
104	Location of forward datum diameter		0.40%	3.2	Gage
105	Location of aft datum diameter		0.40%	3.2	Gage
106	Dimension from forward body holes to inside nose end, min.		0.40%	3.2	Gage
107	True position of holes (3 patterns)		0.40%	3.2	Gage
108	Inside diameter of nose cylinder		0.40%	3.2	Gage
Major B					
131	Wall thickness, aft of inside radius (Check at four random locations)		0.40%	3.2	Gage
132	Wall thickness, forward of inside radius (Check at three random locations)		0.40%	3.2	Gage
133	Thickness through nose end		0.40%	3.2	Gage
134	Any operation missing		0.40%	3.2	Visual
Minor					
201	Finish missing or incomplete		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES: Note 1: Check from nominal radius.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63068 (PA)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.3	Packing (See 5.1)		1 OF 1	N/A
				NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Polyethylene bag missing or improperly assembled		0.65%	3.1	Visual
202	One or more switches missing from any partition		0.65%	3.1	Visual
203	Packing loose, within fiber box		0.65%	3.1	Visual/Manual
204	Fiber box damaged, improperly sealed or marked		0.65%	3.1	Visual
205	Bag damaged, improperly sealed or marked, or missing (Level B)		0.65%	3.1	Visual
206	Packing loose, within wood box (Level B)		0.65%	3.1	Visual/Manual
207	Wood box damaged, improperly sealed or marked (Level B)		0.65%	3.1	Visual
208	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Proving Ground

4.4.3.1.1 Graze Functioning.-A sample of twenty (20) impact switches shall be randomly selected from each lot for this test. Each sample switch shall be assembled onto a cartridge comprised of components (projectile, fuze, shaped-charge liner, etc.) previously tested and accepted in accordance with applicable specifications. If six (6) or more cartridges fail to comply with the applicable requirement, the impact switch lot shall be rejected. If one (1) or more projectiles fail to initially impact the gravel pad, the impact switch lot shall be referred and results of such impacts shall be reported.

4.4.3.1.2 Safety.-All cartridges fired in proving ground testing shall be observed for impact switch separation or breakup. If any impact switch fails to comply with the applicable requirement, the lot shall be rejected.

4.5 Test Method and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspection.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein.

4.5.2 Insulation Resistance.-Minimum insulation resistance shall be tested with the impact switch placed over an electrically conductive mandrel. The mandrel shall have the form and maximum external dimensions (except threads) of an M409 windshield (dwg. 10534412). The mandrel shall include an Insulator Assembly (dwg. 10534417) or facsimile, and a dummy power supply (dwg. 9276238) having maximum envelope outside diameter. The power supply envelope shall be electrically connected to the windshield facsimile, and provision made to electrically connect the test impact switch cup to the test equipment at the dummy power supply forward end. The impact switch being tested shall be held down on the mandrel by a

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15 + 5 pound load on the impact switch nose end. Insulation resistance shall be measured between the mandrel and the impact switch cup after allowing a two (2) second (min.) charging time.

4.5.3 Dimensional Control of Cast, Stamped, or Molded Parts.-In place of the normal sampling associated with the classification of Defects, and subject to approval of the Government procuring agency, the optional sampling plans in paragraphs 4.5.3.1, 4.5.3.2, and 4.5.3.3 may be used for dimensional control of cast, stamped, or molded parts. Use of these optional plans is limited to those dimensions and features controlled by the tool, die, or mold cavity. For all specified tests, visual examinations, and examinations of dimensions or features not developed exclusively by the tool, die, or mold cavity, these optional plans are not applicable. In the event of in-process reject(s) during optional sampling, corrective action applicable to parts produced prior to the rejected, and subsequent to the last acceptable sample, shall require approval of the Government procuring agency.

4.5.3.1 Cast Parts.-A sample of at least five (5) parts (as cast) from each cavity shall be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided. If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be remomed from production. Further, that portion of production since the last control check shall be returned for corrective action. All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above.

4.5.3.2 Stamped Parts.-A sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production. If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production. If any defective parts are found when

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inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned for corrective action. All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above.

4.5.3.3 Molded Parts.-A sample of five (5) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be dimensionally inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned for corrective action. All cavities removed from production because of some fault, may, after reworking, be returned to production, providing they pass the qualification test above.

4.5.4 Ballistic Test.- These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired in a gun for which the cartridge is standard and observation shall be made for proper functioning and for parts separation or breakup in the gun bore or in flight prior to target impact. Smear cameras shall be positioned to record the initial portion of projectile flight as a minimum.

4.5.4.1 Graze Functioning.-The ammunition for this test shall be conditioned at a temperature of 70 Degrees Fahrenheit (°F) ±5 °F for 16 hours min. prior to firing. Test cartridges shall be fired for graze functioning at a gravel impact field, at a range of 1000 meters.

5. PREPARATION FOR DELIVERY

5.1 Preservation & Packing

5.1.1 Level C.-Package each impact switch nose down in a .002 inch thick, Type II, any grade, Finish 1, Spec L-P-378 polyethylene film bag. Size of bag to be 10. inches open face x 22.

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inches deep. The bag may be tapered at the bottom to coincide with the contour of the impact switch. Switches shall be inserted nose down in the bag and the excess material at the top shall be inserted within the switch. Pack 30 bagged switches, within half-slotted partitions, in a Type CF, domestic class, variety DW, 275 grade, RSC style fiber box Spec PPP-B-636. The inside dimensions of fiber box to be 18 3/8 x 12 1/4 x 18 1/4 high. The overall dimensions of the half-slotted partitions shall be 18 1/4 x 12 1/8 x 17 high. Each cell of the partitions shall be 6 inches square x 17 high. The material used for the partitions shall be of same material as fiber box except variety SW may be used. Pack five switches nose up in each of six compartments. Add a filler of the same material as the fiber box between the inner and outer flaps on bottom and top of fiber box. Tape fiber box in accordance with Spec PPP-B-636 as applicable for boxes overpacked or shipped as exterior containers.

5.1.2 Level B.-In addition to fiber box specified in Level C, overpack fiber box in a Class E, Spec MIL-B-117 bag after blunting corners of box. Exhaust all excess air within bag and seal bag in accordance with Spec MIL-P-116. Size of bag to be 32 min. open face x 34 min. deep. Pack 12 bagged fiber boxes (3 x 2 x 2 high) in an overseas type wood cleated-plywood box in accordance with Spec PPP-B-601. Box to be provided with skids in accordance with Para. 3.3.3 of Spec. Requirements for box to be for 440 pound contents (Ref Table III). Inside dimensions of box to be 38 1/2 x 37 3/4 x 38 3/4 high. Add filler material 36 x 36 x 1/16 thick on ends, sides and top as required for tight packing. Filler material to comply with Spec MIL-B-3106 or MIL-F-50449.

5.2 Marking

5.2.1 Level C.-Marking of fiber box to be in accordance with MIL-STD-1169.

5.2.2 Level B.-Marking of fiber box, bag, and box to be in accordance with Level A, MIL-STD-1169.

6. NOTES

6.1 Intended Use.-The part covered by this specification is intended for use as part of the M539, PIBD, Fuzing System.

6.2 Ordering Data.-See MIL-A-48078 (PA).

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6.3 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078 (PA). Submit equipment designs, as required, to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-T-T
Dover, New Jersey 07801

6.4 Inspection Lot Size.-It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of test or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Proving Ground Test Summary.-

Test	Sample Size	Requirement	Test Procedures Paragraph
Graze Functioning	20	3.4.1	4.5.4
Safety		3.4.2	4.5.4

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A119

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

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